Forside

Eksamensinformation
ASDK20007E - Digital Methods

Besvarelsen afleveres af
Oliver Fredborg Smietana
kph383@alumni.ku.dk

Arvin Ammary
hjn900@alumni.ku.dk

Aleksandar Yuri Valev
wtg297@alumni.ku.dk

Nikolai Sonne Steenberg
xmf434@alumni.ku.dk

Eksamensadministratorer

Bedømmere
Kristoffer Langkjær Albris
Eksaminator
kristoffer.albris@sodas.ku.dk
+4535320485

Hjalmar Alexander Bang Carlsen
Eksaminator
hc@soc.ku.dk
+4535334449

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How Ukrainian Military Brigades Mobilize Support Through Social Media

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Course Administrators: Kristoffer Langkjær Albris & Hjalmar Alexander Bang Carlsen

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Authors:

Arvin Ammary (Exam number: 4)

Aleksandar Valev (Exam number: 63)

Oliver Smietana (Exam number: 47)

Nikolai Steenberg (Exam number: 13)

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Introduction (All)

On the 24th of February 2022, Russia launched an invasion on Ukraine after years of geopolitical tension. This invasion has resulted in an ongoing war that has cost hundreds of thousands of lives and the occupation of large parts of Ukrainian territory by the Russian military (War.Ukraine, 2024). While the Ukrainian forces have been able to push back on Russian advances, donations from Western nations come with delays and unreliability (Méheut, 2023), and the Ukrainian government has been struggling with recruitment of military personnel (Koshiw, 2024).

The Ukrainian military have therefore been seeking help through alternative means, namely through recruitment, donations and support directly from the Ukrainian population. The individual military brigades of the Ukrainian army have been advertising themselves and their activities on various social media platforms, including YouTube, Twitter, and Telegram (Arraf, 2022). With the growth of social media platforms like Telegram, real-time updates and footage from individual military brigades are now accessible to the public during wartime. In Ukraine, military brigades now have their own social media managers, websites, graphic designers, and more, all with the goal of garnering support, through recruitments and donations from private citizens (Méheut et al., 2024).

This leads us to the following research question of interest: *How do Ukrainian military brigades mobilize support from Ukrainians through social media?*

Through a combination of qualitative and quantitative social data science methods, we are able to investigate this research question through digital online ethnography, machine learning, and computer-assisted topic modeling.

We will begin this paper by going over the theoretical framework of framing analysis and how it can be used to analyze how the military brigades frame their messaging for the purpose of mobilizing support. Then, we will introduce the methodological framework of the analysis, which will include a digital ethnographic approach to the investigation of the military brigades and the messaging service Telegram, the chosen data site for this paper. We will further cover the data collection of this data site and how we use qualitative coding to facilitate our computer-assisted analyses of specific topics pertaining to our research question. The qualitative approach to our framing analysis and the quantitative approach to

our analysis of the message frequency and engagement of these topics will be outlined next. Lastly, we will cover our results from these analyses and discuss methodological limitations and possible alternative approaches to our research question.

Theory (Alex)

Our qualitative analysis will be based on the theory of frame analysis, originally developed by Erving Goffman (1972). Frames were conceived of by Goffman as ways in which people structure experiences and make sense of reality. As such, they can serve as a basis for further individual or collective actions in a given context. In this sense, frames provide social scientists with a robust scheme for understanding social actors' perceptions and actions. While broad in scope at first, this approach has proven to be foundational for the development of multiple fields of research, such as behavioral economics (Kahneman, 2013), media studies (Hambor, 2023), and social movement participation (Daniel et al., 2023).

A lot of literature has been established in said fields since Goffman first coined the term "frame analysis", further building upon the theory. For the purposes of our analysis, we will be referring to concepts outlined in the article "Frame Alignment Processes, Micromobilization, and Movement Participation" (Snow et al., 1986).

In it, the authors aim to bring more awareness to the confluence of social psychological and organizational circumstances at play when people decide to back or take part in social movement organizations (SMOs), as well as some overlooked characteristics of social movement participation, such as temporality. To do so, they introduce the term frame alignment together with four frame alignment processes. Two conditions need to be met for frame alignment to occur. First, the values and principles in which an individual or a group believe in have to coincide with those of an SMO and the social cause they are advocating or fighting for. Second, the potential new supporters must come to think of the social issue at hand as a clear case of injustice, which can be rectified, thus assimilating an injustice frame.

The three frame alignment processes represent different mechanisms, or micromobilization tasks, that SMOs use in order to achieve frame alignment with potential supporters.

Frame amplification refers to an SMO's attempt to clarify its purpose or to inspire and renew interest in the cause either through value amplification or belief amplification. The need for

frame amplification can vary widely, from hopelessness and emotional fatigue in supporters, to strict restrictions from the government. When the speaker of a social movement reiterates the importance of a value like freedom, it is called value amplification. On the other hand, if the same speaker is reminding despondent constituents of the belief that the social cause is feasible and there is still hope for it to succeed, this could be thought of as belief amplification.

Frame extension involves the expansion of an SMO's framing with the expectation to attract more potential supporters, who may otherwise be uninterested in participating.

Finally, frame transformation entails the reframing of an SMO, again with the prospect of gaining more supporters than otherwise. For example, frame transformation may be needed when the movement is advocating for causes or actions which go against conventions and social norms.

Frame alignment has served as the basis for many research efforts in the field of social movements. Although the context of war is different from a regular social movement or protest, the common thread between the three contexts is the social cause at the heart of them. Therefore, we are inclined to think that frame analysis and frame alignment theory will also be suitable for our study.

Given the context of our research case, we can expect to mainly encounter messages that would fall under frame amplification, since Ukrainians are most likely aware of the existence of the military units fighting against Russia and already want their country to win the war, but brigades might need to rekindle their enthusiasm. Overall, the theoretical framework of frame alignment provides us with the tools necessary to investigate how Ukrainian military units attempt to mobilize support from civilians.

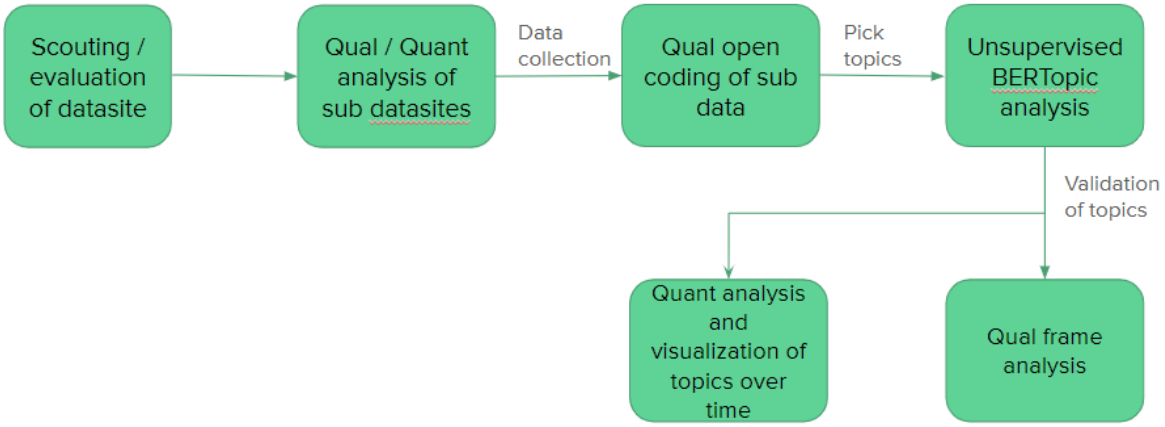
Methodology

Sequential Analysis Design (Arvin)

The methodology we will use to answer our research question will follow a sequential analysis design. This will include sequential steps for the analysis that uses either qualitative, quantitative, or mixed-method approaches at each step of the process. The different methods will be used both confirmatory to validate patterns found within the methods and complementary to investigate different aspects of our research question. As Dominguez and Hollstein (2014) discuss, mixed methods can bolster the reliability of research, through data triangulation while the use of multiple methods can deepen the analytical insights. This is especially important for our data source, since it is highly contextual and given our previous unfamiliarity with the Ukrainian context, orienting us qualitatively is a crucial prerequisite for quantitative work.

The first step in our analysis is a qualitative netnographic scouting and evaluation of data sites that are suitable to investigate our research question. This step is crucial to ensure that the data we collect is fitting for our case and representative of the entities we are studying. The next step consists of our sampling strategy of the chosen datasite. As we will describe in the following sections, our chosen datasite, Telegram, consists of sub-data sites, as each military brigade has its own channel. Through qualitative and quantitative means, we will evaluate both which channels and which messages in the channels to include in our analysis. Having collected our data, we will conduct a qualitative open coding of a sub-sample of the data. This exploratory approach has the purpose of getting knowledge about the themes and topics that occur in the messages. After picking our topics through a more focused approach, we will use an unsupervised topic modeling technique to extract important topics and keywords from the messages. We will use this for the validation of the keywords and topics found through our own qualitative open coding of the messages. Once we have gathered our chosen messages and topics related to these messages, we will conduct a qualitative analysis of the frames used by the military brigades within the different topics. Furthermore, we will conduct a quantitative analysis and visualization of how the frequency of the topics has changed over time, as well as how the online engagement among the users differs across the topics.

Figure 1: Illustration of our analysis design



Netnography (Alex)

In this section, we will conduct an iterative netnographic approach to the investigation of the relevant actors of our research question, namely Ukrainians and Ukrainian military brigades. The approach relates to what Kozinets notes about online fields and how researchers should develop an understanding of the research site in order to grasp the dynamics involved (Kozinets, 2019).

Throughout our netnographic campaign, we will systematically record our observations and reflections of our data exploration in an immersion journal (See appendix). We will include descriptions and interpretations of our observations and any changes we make based on our findings. Furthermore, we will be keeping track of the changes we have made to our approach to the research topic based on the feedback we get on our proposed decisions.
Due to our research question pertaining to Ukrainian military brigades who likely have different ways of using social media, and certain colloquialisms that are less known outside their circles, our netnographic approach ensures that we get a deeper understanding of the phrases they use, what they mean, and how they interact on social media.

Datasite (Arvin)

A 2023 study has estimated that 72 percent of the Ukrainian population uses Telegram as their main source of news (Palikot, 2023). The popularity of Telegram skyrocketed after the

Russian invasion in 2022 and it has now become the most popular platform for news among Ukrainians, surpassing online news websites, television, and other social media sites like Facebook and YouTube (Palikot, 2023). The broad usage of Telegram among the population and military units, combined with the popularity spike related to the Russian invasion makes the platform a fitting datasite to investigate.

Telegram is a messaging service which was founded in Russia in 2013. Telegram features channels that can be either public, where any user can search for and join the channel, and private, where the channel is not able to be looked up through Telegram's search function and only users with an invite link can join. Telegram allows for differing levels of user interaction where some channels only allow replying and reacting to messages from the channel moderators, while other channels have a flat structure where everyone can send messages freely. Furthermore, a wide range of media types is permitted, including, text messages, images, videos, GIFs, stickers, and embedded websites.

Some of the key aspects of Telegram are its message encryption, evasion from governmental intervention, and lack of content moderation which has drawn attention from users who want to access content that gets filtered out on more mainstream social media platforms (Molla, 2021). For this reason, Telegram has been subject to much controversy since its creation with multiple countries banning the platform (Kilpatrick, 2015). Meanwhile, the Ukrainian government too has considered banning Telegram due to worries of pro-Russian channels spreading misinformation regarding the war (Palikot, 2023).

Data Collection (Nikolai)

This section will describe our sample, the underlying sample strategy and the methodology used for collecting, filtering and storing the data.

The focus of our study is on the Ukrainian military, specifically brigades and the individual battalions of which brigades are composed of. In a US context, a brigade refers to around 5000 soldiers, while a battalion typically contains 1000 soldiers (Military Units, n.d.). These numbers are likely not directly applicable to Ukraine, but they can serve as a rough estimate. For the purposes of this study, we are interested in sampling any Ukrainian Brigade or

Battalion, now referred to as military units. This is then a strategy of sampling from known actors.

In our data collection we first had to obtain a list of Telegram usernames from the known sample of Ukrainian military units, given their presence on Telegram. To do this we manually went through all military units on a site that tracks the structure of the Ukrainian military (Ground Forces of Ukraine, 2022). For each military unit we included the Telegram username if it was listed on the site, otherwise we tried finding the username on other social media links. If unsuccessful, all available social media unit handles, as well as the name of the military unit in English and Ukrainian, were used as search strings on Telegram. We then collected all channel messages including media captions, and their emoji reactions for each military unit in the list, in the period from the start of the full scale war in 2022.02.24 to 2024.05.04. This was done using PRAW, a python wrapper for the Telegram API (Praw, 2023). Messages and emoji reactions were collected in and stored in a SQLite database to ensure data integrity. The collected meta-data included message type, date, edit date, message views, message forwards, chat name from which a message is forwarded and channel member count. To improve our sample we extracted mentions and links to other channels from the message data. Here we manually identified channels that we missed in our earlier sampling step, and ensured to collect the same data from them. Then we repeated this process once more, as a form of snowball sampling.

In regards to GDPR, we are collecting information from public sources, and not specific private individuals. However, personal information may appear in the data, especially in the form of names in messages commemorating fallen soldiers or portraits of soldiers. While the latter is considered personal information under GDPR, this does not extend to information about deceased individuals. Further, the data subjects are located in Ukraine, and are thus not within the EU. This then does not necessitate the implementation of specific precautions for data processing and storage. To accommodate ethical considerations we will not share our data, and redact names from quotes.

After the data collection, we filtered channels with fewer than 300 posts to exclude low-engagement and new channels, resulting in 33 channels. These limits are chosen as we are interested in the strategies brigades use and the mobilization from these, and the strategies can only be analyzed through a significant number of posts. Since all messages were in

Ukrainian, they were translated into English before the analysis using Google translate API. Due to API limitations, an insignificant number of messages above 3700 characters in length were filtered out. To check translation quality, some messages were read and compared to translations from other translation services such as DeepL. The translations from Google translate seemed to be coherent, readable and comparable to DeepL translations. Further, in the context of political science, machine translations compared to human translations have also been found to be highly similar (De Vries et al., 2018). After translation, the dataset consisted of 52173 messages.

Qualitative approach

Coding (Arvin)

To first get a better understanding of our observations, we conducted an open coding of a selection of the observations from our data collection. We sample the ten most viewed messages within each of the 33 brigades to get an overall representative sample of the Ukrainian military brigades. These messages were manually read through in an explorative manner to discover any themes that could be of analytical interests across the different brigades' channels. Certain topics, keywords, and colloquialisms were noted down and researched to get an understanding of their meanings and contextual implications. After having gotten an extensive idea of the themes within these Telegram channels, we conducted a more focused coding of the message samples. This consisted of creating a list of topics that reoccurred in the messages, which we found to be 13, and coding the messages as belonging to one or more of the topics. This left us with an idea of which topics that are common across the different channels and how these topics are presented. We decided to focus on five distinct message topics for the analysis of our research question, which we created a closed codebook for, detailing a clear coding scheme (see appendix). These topics are *Recruitment, Military Success, Heroism/Motivation, Donations,* and *Russian Hate*.

BERTtopic analysis (Nikolai)

BERTopic is a topic model that operates on contextual document embeddings, and it is also performant on short documents, making it relevant for our textual data (Carlsen & Ralund, 2022). We used the all-mpnet-base-v2 model to create document embeddings, which were

then reduced in dimensionality using Uniform Manifold Approximation and Projection (UMAP). The embeddings were then clustered with the HDBSCAN algorithm, after which the documents were tokenized with a custom CountVectorizer that removed uninformative words, including stopwords, military unit names, and names of social media sites. Then class-based TF-IDF was used to create the topic representation, which was refined by applying KeyBertInspired to identify relevant keywords. Notably, this process was iterative as recommended by Carlsen & Ralund, (2022), meaning that parameters for HDBSCAN, UMAP, and CountVectorizer were iteratively adjusted to ensure topic distinctiveness while keeping the number of topics manageable.

The final model included around 70 topics in total. Based on the keywords and the example document for each of the model topics, we assigned topics to one of our five existing qualitative categories based on the closed codebook. This can be seen as a type of computer-assisted approach, which according to Carlsen and Ralund (2022) is crucial in identifying patterns not captured directly by the model. The BERTopic model was then updated by merging all topics assigned to the same qualitative topic, after which new document probabilities were calculated. Topics not included in the merged topics were dropped, resulting in a corpus of 8462 documents distributed between the five topics. To validate the coherence of the merged topics with the qualitative topic descriptions, we used stratified sampling to randomly sample 30 documents from each topic with a topic probability > 0.50. Each document was then carefully read through and marked as “relevant” or “not relevant” to its assigned topic. The proportion of relevant documents within each topic can be seen in Table 1.

Table 1: Topic validation

Topic	proportion	Topic_name
0	0.833333	Motivation/Heroism
2	0.700000	Russia Hate
3	0.733333	Military Success
4	0.900000	Donations
5	0.733333	Recruitment

After arriving on a final set of topics through combining the results of our qualitative coding with the BERTopic model, we were able to extract documents for the frame analysis. We used all topic relevant documents from the validation sample described earlier, which provided us with comprehensive representations of each topic. We will proceed with identifying what types of frames and frame alignment processes are at play in the different topics that we found in the messages of the military units. To do this, we will do a close reading of the sampled documents by topic and interpret their content according to the frame alignment theory. We will also add example documents which serve to showcase and substantiate our claims.

Quantitative approach (Oliver)

To further explore our research objective, we will support our qualitative analysis with a quantitative analysis. The purpose of this analysis is to enrich our framing analysis of individual posts and authors with knowledge on the temporality and reception of the topics. This quantitative analysis will involve checking the prevalence of each topic over time within the specified period and finding a measurement for the level of interaction with a given post.

The three possible engagement metrics on a post on Telegram are the number of views the post has, the number and type of emoji reactions, and the number of comments, including a close reading of the comments. We believe that the first two metrics—views and emojis—are valuable data sources to approach our research objective quantitatively, as it is informative of how and how much viewers engage with the content, while still being easily quantifiable.

The first analysis, which involves tracking the prevalence of topics over time, can be done using the BERTopic model that can measure the frequency of posts on a particular topic over time. The quantitative insight is further enhanced by comparing it with specific periods of the war. Once we know that a particular topic is popular during a specific event, it is possible to evaluate the framing in direct response to such an event, and thus give us a better understanding of the reasoning behind the military brigades' behavior and activities in their Telegram channels.

In Matamoros-Fernández (2018), the authors explore how emojis are used in right-wing Belgian discourse, particularly how angry emojis are used to incite racism against Muslims. This serves as an example of how reactions can be analyzed through the emoji reactions to posts. We continue with this idea and explore how quantifying emoji reactions can bring insights into how posts are received. In order to understand the meaning of the emojis, it is important to understand each topic, and have a solid understanding of the context they are posted in. That is because, while some emojis are intuitive emotional reactions to a post, such as a vomit emoji to news about muslims on racist sites such as in Matamoros-Fernández (2018), other emojis can be used in a unique way in the given context, or they can be highly sarcastic reactions. Through our qualitative framing analysis we establish the necessary understanding and knowledge of these possible scenarios in an Ukrainian context, and thus an analysis of the emojis can provide valuable insights.

The second analysis will be to use emojis as a measure of reaction sentiment for messages in each topic, with the purpose of understanding how the communication is received across topics. Here a dictionary approach will be used to quantify emoji sentiment for each topic. To ensure alignment with the data, a custom dictionary is used to classify the most popular emojis from each topic. The emojis "🤬", "💔" and "😢" are classified as negative while the remaining emojis are classified as positive ("❤️, 👍,❤️‍🔥,🔥,🫡,🙏,👏,😁,🤣,💯").

For the third quantitative analysis, we are interested in assessing whether Ukrainians were indeed mobilized, as posed in our research question. Given data limitations on donations and recruitment, we will use the average number of emoji reactions per view for each message, aggregated by topic as a reasonable gauge. This measure is similar to engagement rates associated with marketing, however we will use the term "reaction rates" since we do not account for the number of Telegram followers in the calculation.

Our research question is about how the brigades mobilize the people, and gauging the views and emoji reactions can help answer this question in the sense that we implicitly ask for the successful mobilization. The successful mobilization attempts online will likely also correlate with what mobilizes people in the real world. We also consider the posters to be aware and expect the reactions of the viewers. This means that if the brigades get an angry smiley emoji to posts about Russians, or if they get a heart emoji to posts about Ukraine, we believe this was the posters intent.

Analysis

Qualitative analysis (Alex)

We have conducted a close reading of a large sample of the posts within our 5 chosen topics. While reading these we analyzed which frames the Ukrainian brigades used and paid close attention to which frame alignment processes they have used to strengthen their attempts at mobilizing the Ukrainian people. The general frame of the Ukrainian brigades is of course related to them being a "good" brigade that fights "evil", and the Social Movement in their posts is always the war against Russia. However, the exact way in which the brigade is "good" varies between contexts and is revealed through our analysis.

Heroism (Oliver)

The brigades tell the stories of individual soldiers' life and death as a way to strengthen their reputation as a valiant military that fights for Ukraine, with the ultimate consequence that Ukranians should support and idealize the military. Martyrdom and the benefit of showing dead fighters in a good light outweighs preemptions of any critique of losing soldiers. The brigades thus, use frame amplification to turn the death of a soldier into an epic story of a hero.

> "[Soldier Name], born on [Date]. Was under contract since 2019. Then on May 28, 2023, he voluntarily mobilized to the front. On [Date], while performing a combat mission to destroy enemy positions as part of a group of flamethrowers, he personally pulled out an injured comrade from under enemy fire, thereby saving his life. [Soldier Name] died on [Date], near the settlement of [City Name], Donetsk region. 59 separate motorized infantry brigade. Please sign the petition on awarding the honorary title of Hero of Ukraine (posthumously) to a serviceman!"

This brigade has lost a soldier, and the post is about his achievements and a call to award him a title. This brigade, and many others when they write about a soldier's death, does not write defensively or introspectively about it. It is also easy to imagine that these stories are either specifically chosen because they make the perfect story of a hero, and/or they are embellished and uplifted to become symbolic of the bravery and sacrifice of the Ukrainian soldiers. This

is a good example of how the Ukrainian brigades use frame amplification with individual stories in order to boost their frame. In another post a brigade writes:

> "[Soldier Name] died heroically while performing a combat mission…. The war continues to take away the best sons of our Motherland, so we have no right to stop, lest the cause of the fallen heroes be in vain."

First, the soldier died "heroically" and not tragically or unfortunately further substantiating our claims that the brigades are not concerned with communicating introspectively about how they can reduce the deaths. Secondly, this brigade goes a step further by not just glorifying an individual person who has died, but by saying the best sons have died they approach a glorification of dying, and they also use a form of guilt for all the soldiers who have already died, in order to mobilize the next soldiers. This can be seen primarily as another example of frame amplification but also as a form of frame extension as the post hints towards a negative view of not continuing that is less about the heroism of soldiers and more about the shame of not serving or wanting to stop the war. The post thus extends the frame of a military that fights for Ukraine, by leveraging anger against those who wish to stop the war or not fight in the military.

Russia Hate (Alex)

The brigades use examples from the war of how the Russians behave wrong, dishonorably or otherwise immoral to frame themselves in opposition to these qualities, as a force of righteousness and justice. Again, we discover that the Ukrainian brigades use frame amplification to strengthen their frame, but there are also attempts at a complete frame transformation in order to place the brigades and military as a bulwark against crimes against humanity. This frame amplification is illustrated by a post:

> "Meet – [Name, picture and home city of Russian soldier]. During the hostilities in the village [Village name, region name], the occupier stole a mobile phone from the girl…. The payback will come, we will find everyone. Glory to the Nation Death to enemies."

Here the brigade uses the immoral behavior by a Russian soldier and places themselves as the force that will provide retribution for these acts. The unjust feelings are amplified by the

example being a Russian soldier stealing from a Ukrainian "girl", which is another case of picking the best examples for their point. Thus they appeal to the feelings of unfairness that a reader is left with when they hear about the behavior by Russians, and use this to mobilize enlistments or civilian support of the military.
In the following snippet the brigade attempts a frame transformation of the war:

> "Orcs have nothing to oppose in a fair fight. They hit you on the sly with rockets and mopeds. Their goal is to leave the civilian population without light and heat in winter. In such conditions, we have every right to execute the bastards who are committing genocide of the Ukrainian population."

The brigade is attempting to transform the framing of themselves and the war from a defensive war, into a defense against genocide. They use these examples to stoke a feeling of anger against Russians and a feeling that they are not only attacking Ukraine but they are also doing it with the intent to kill civilian Ukranians. The resulting idea is that in some way the civilians are already in the fight against Russia just by being Ukrainans. In addition to this the snippet contains further amplification of the frame of just and honest Ukrainian military against an uncivilized, destructive and immoral "orcs" fighting with barbaric means.

Military Success (Oliver)

The sample of documents from the "military success" topic mainly consists of reports on the efficacy of a given element of the Ukrainian military force. This element can range from a whole military unit to a piece of equipment or weaponry, like a drone. More specifically, the messages always contain concrete numbers when, for example, describing how many Russians bases have been eliminated by a military unit. In this sense, these documents are framed as reports, updating the reader on how well the military activities are going with a very quantitative approach. Additionally, it is very common for a message to end with "Glory to Ukraine!". An example message reads:

> "New prey on the account of our lions. This time, operators of strike drones of the 103rd brigade hunted down a T-80 tank and its commander, who was trying to back up. Such a fate will befall all those who encroach on our will! Glory to Ukraine!"

Referring to Snow et al. (1986), we interpret these reports as belief amplification, a branch of frame amplification, which serves to reinforce the belief that victory is one step closer with each successful operation. Therefore, we argue that messages like this can rally support and mobilize the reader to donate or apply to be part of a battalion.

Donations (Alex)

Shifting focus to the "donations" topic, the sample contains messages with calls for help, asking for a variety of resources. Often, documents include a description of how the resources will be allocated and utilized effectively. The documents are characterized by a high sense of urgency:

> "DEVIATION FROM PLANS Our fighters need help again! We wish you good health. We are asking for help. "93 Brigade Cold Ravine". A thermal imager is very necessary. The goal is 42 thousand hryvnias!!!"

While the documents in this topic do not evoke an already established frame alignment process, we would say they resemble the process of frame amplification. Typically, the two frame amplification processes are value amplification and belief amplification, but we would argue these messages can be thought of as a sort of utility amplification, emphasizing the usefulness of the requested resources and how they would serve to help soldiers or civilians.

However, a lot of the posts on the donations topic are somewhat counter to the frame. Where, in Military Success and in Heroism posts, the frame was clearly about the Ukrainian military being glorious, strong and fighting for good, in donations the brigades have to perform a form of confession to inadequacy in order to even ask for the donations. Why is a glorious and experienced military asking for thermal underwear? This confession can clash with an idea of a glorious, heroic military, but the frame could also possibly be extended with a David vs Goliath symbolism. This is a characterization that many Western politicians have used, for example EU's foreign policy chief, Josep Borrell: "Ukraine will succumb to the invading Russian forces "in a matter of days" without military support from Western countries" (Euronews, 2023).

In this quote it is very clear that his view is that the Ukranians are fighting a much stronger opponent. It is noteworthy that while there are posts asking for help and donations, there is

not an attempt to extend the frame to include these posts. Instead they appear to be primarily based on urgency as above or community and support for troops as in this example: “Dear volunteers! We are very grateful to you for your great support and constant help. Together we are a powerful force that accelerates and brings our common Victory closer!”, while not acknowledging the power difference between the two militaries.

Recruitment (Oliver)

When it comes to direct recruitment posts the brigades focus on empowering rhetoric encouraging the civilians to take action and join the forces. Here the brigades use frame amplification and frame extension. This is also done with a hint of shame or negativity surrounding not taking action. The backdrop of these calls to actions is the reputation that the military has built up with the Heroism, Russia hate and success stories. Recruitment is a big part of the reason they have a social media presence to begin with and in this topic they are capitalizing on the goodwill they have built up:

> ”The 12th "Azov" brigade continues to fight. Our army is increasing. And we invite you to join our ranks. We do not wait for a better future, but fight for it. We do not look for flaws, but change ourselves. This is our choice. Make your choice and you. Fill out the questionnaire on the website azov.org.ua.”

Here, the Azov brigade tries to mobilize people to join their brigade by illustrating that they have a choice. They appeal to all the people worrying about the future, and how the war might turn out, and try to use those feelings to say that the civilians should make a choice to do something about it. The brigades try to extend the frame for the military to include more people by using empowerment and self-improvement rhetoric. They also hint at acceptance of different people by saying “We do not look for flaws”. All of this frames the military as actionable and successful and a place where you can improve yourself.

In another post by another brigade the pressure to make a choice is even larger:

> ”… Do not wait - direct your energy in the right direction! Prove to everyone that you are a true patriot! Learn to win, victory is near!...”.

Here the brigade is saying the reader should prove themselves to be a true patriot by joining the brigade. This framing fits together with the heroism and martyrdom framework as an

amplification. As the Ukrainian military enjoys large support from the population of Ukraine, which is natural seeing as they are under attack, the military enjoys large power in defining what patriotism is and who is a good Ukrainian. Furthermore, with words such as “right”, “patriot” and “victory”, they activate the ideas that are generated in the three topics of military success, russia hate, and heroism, and leverage those to recruit people into their brigade.

Topic Interactions (Oliver)

Based on these findings, we can categorize the three topics military success, heroism, and Russia hate as “construction topics”, because this is where the military reputation is constructed and the two topics of recruitment and donations are categorized as “consumption topics”, because the reputation is used and consumed with these topics. The brigades build up an idea of a great, glorious, experienced and technically capable military, and then this idea is used to recruit more soldiers into the brigade and to solicit donations from civilians. However, the idea is weighed down or limited by the posts asking for donations.

This mechanic that applies to brigades can be compared to a mechanic that applies to social media influencers. The influencers must strike the right balance between original content and advertisement, and they must also be creative in how they incorporate the advertisements into their channel, lest the followers feel they have “cashed out”, that is, they are too obvious about or have too many advertisements. This process relates to the concept of idiosyncrasy credit (Hollander, 1958), which refers to the capacity for an individual to display behavior which deviates from the expectations of the group they belong to. When an individual behaves according to the accepted group norms, they receive idiosyncrasy credit, but deviancy from expectations leads to reduction of said credit.

This idea can be applied to the way we interpret the dynamics of the brigades’ Telegram channels. The reputation that the brigades build up is exchanged for recruitment or consumed by the posts asking for donations. The brigades may be happy with the situation as it gives them donations and therefore makes a direct difference on the battlefield. But not accounting for the realities on the battlefield in the frame may also be limiting the mobilization of the Ukrainian people as it can create cognitive dissonance as people try to align the two

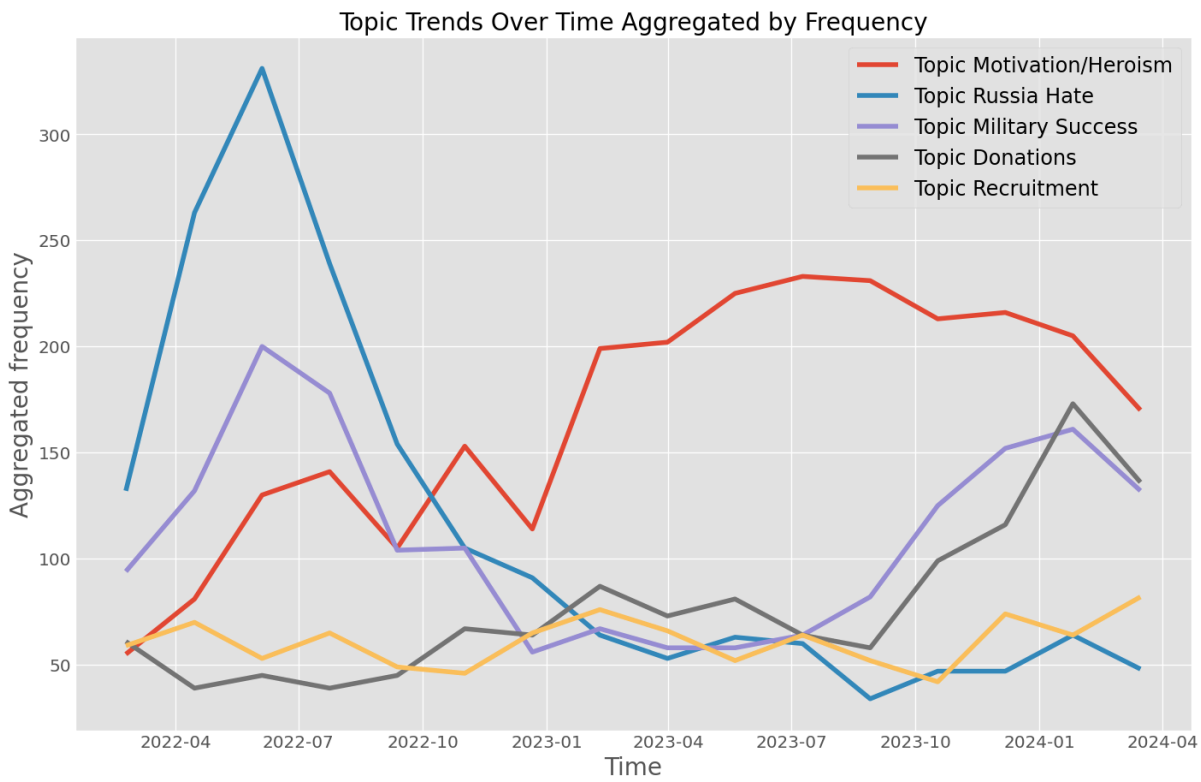
conflicting ideas themselves. The brigades are thus performing a balancing act when they ask for donations, between being someone who is admired and someone who needs help.

Quantitative analysis

Topic frequency over time (Arvin)

This part of the analysis will focus on how the prevalence of the five chosen topics have changed over the course of the war. The results of this quantitative analysis allow us to discover certain changes in how the Ukrainian military brigades communicate about the war to the Ukrainian population by focusing on how the brigades "construct" their image and how they "consume" it. The topic trends over time have been visualized in Figure 2.

Figure 2: Topic trends over time aggregated by frequency



The first topic we will focus on is "Russia hate" which pertains to negative sentiments towards Russians. The figure shows that this topic has the largest overall change in frequency over the course of the war, with a sharp increase in the first months of the war, being the most popular topic by far. Newspaper reports have shown that anti-Russian sentiment indeed did increase shortly after the invasion, not just in Ukraine (Petrova, 2023) but in much of the

Western world (WION, 2022). These messages could therefore have had the goal of garnering support among the population by feeding into their anti-Russian sentiments. In other words, This was an initial way to "construct" the image of the military. However, the number of messages with anti-Russian sentiment quickly declined in the latter half of 2022 and has steadily declined since then, becoming the least frequent message topic among the five chosen topics. Despite this decline, anti-Russian sentiment remains record-high in Ukraine with 89% of the population having a negative perception of Russia as of 2024 (DPI, 2024).

The topic "Military Success" pertains to messages showing the brigades' victories in the battlefield. This topic has seen large changes throughout the war which could be due to changes in the military gains and losses on the Ukrainian side. The topic trend quickly increased after the invasion and peaked in around mid-2022.
This initial period of the war saw Russia capture large parts of Ukraine, with Russian forces reaching the capital Kiev shortly after the invasion (Jamali et al., 2022). By April of 2022, Ukrainian forces were able to regain much of their lost territory (Singh et al., 2023), which explains the increase in military success messages during this period. However, the second half of 2022 saw a large decrease in messages on this topic. This is unexpected, as this period of the war was represented by Ukraine's successful counteroffensives, which saw them regaining large amounts of territory (Sengupta, 2022). Yet, the topic trend saw another increase in the second half of 2023, however this period of the war was characterized by Ukraine's second major counteroffensive, which saw them struggling to regain territory and was ultimately deemed a failure (Page, 2023). It is therefore possible that the use of "military success" messages by the brigades is to garner national support, and in our words "construct" the image of the brigades, to gather donations during periods of the war where the brigades might be struggling.

This argument is further supported by the "Donations" topic trend which has followed the same increase in frequency in the second half of 2023. As the qualitative frame analysis of the messages showed, donation messages often include descriptions of how the resources will be allocated and utilized effectively, but it also can "consume" the image by damaging it because it does not fit the frame. It is therefore likely that military success posts are used in conjunction with posts asking for donations to incentivize people to donate by showing the military successes caused by people's donations.

The trend of “Military Success” and “Russia Hate” topics stands in contrast to the “Motivation/Heroism” topic, which has been the most popular topic since the end of 2022. With the military success topic trending downward at the end of 2022, the Motivation/Heroism trend followed an almost inverse trajectory, which continued to grow through the first half of 2023, while military success messages declined. Motivational messages and messages about heroism dominating the Telegram channels during this time seems to line up with the period of the war that has been considered the “stalemate period" (Sestanovich, 2024). This can be seen as a change in strategy for how the brigades “construct” the image. An explanation for the motivational messages dominating this period could be that the military brigades do not have much success to show to the public, nor an urgent need for donations or new recruits. The qualitative frame analysis showed that these messages often tell the stories of individual soldiers’ lives and deaths in heroic ways to strengthen their reputation. It is therefore likely that the military brigades resort to these stories and posts to keep support from the public up during slow periods of the war. This further explains the recent stagnation and slight decline in the topic trend after the recent Ukrainian counteroffensive.

The “Recruitment” message topic has been among the lowest topics with small fluctuations throughout the war. However, since the end of 2023, messages pertaining to this topic have been increasing, and at the time of the data collection, have reached the highest they have been throughout the war. This recent increase in recruitment messages points to a larger demand for troops within the military brigades, and the brigades subsequently focus more directly on “consuming” their image by posting more about recruiting. An inspection into this trend shows that while many Ukrainians volunteered to defend the country at the start of the war, military losses and a decrease in morale has recently resulted in a large need for more military personnel (Koshiw, 2024). Some military brigades have expressed dissatisfaction with the Ukrainian government's ineffective attempts to recruit military personnel, which has resulted in the brigades attempting to advertise their brigades for recruitment by themselves (Peleschuk, 2024).

Emoji interactions (Nikolai)

This part of the quantitative analysis will focus on using the emoji reactions associated with the Telegram messages to analyze how topic frames are received by Ukrainian readers.

In the dataset, there are a total of 73 unique emoji types in the reaction data of which many are unique and thus hindering comparison across topics. Instead the 10 most popular emoji reactions for each topic are chosen (see below). Generally, there is a significant overlap in the emojis used between the topics and the frequency of usage. It is however notable that for some topics, specifically "donations" and "Russia hate", the angry swearing face "🤬" and the sad face "😢" are quite frequent. Based on earlier qualitative readings, these are likely used to express negative sentiment directed at the message content, not the unit. The rest of the emojis seem to be expressions of positive sentiment, which can be interpreted as support. The most popular unique emojis are: ❤️👍😢❤️‍🔥🔥🫡💔🙏👏🤬😁🤣💯

The emojis can be analyzed quantitatively in terms of their sentiment by classifying them as negative, neutral or positive. The counts of positive and negative reactions can be seen in Figure 3. The figure indicates that direct calls for recruitment receive a high amount of positive reactions, and relatively few negative reactions. Direct calls for donations do not receive a high number of positive reactions compared to the indirect mobilizations topics. Importantly, most of the negative sentiment is likely not directed towards the support mobilization effort, but rather towards some of the frames used within each topic in order to motivate donations and enlistment. Keeping this in mind, the topics "Donations", "Motivation/Heroism" and "Recruitment" have the highest rate of negative reactions, hinting at a framing technique of leveraging negative reactions and outrage to mobilize support. This is opposed to the positive reactions used in "Military Success".

Figure 3: Reaction sentiment distribution per topic name

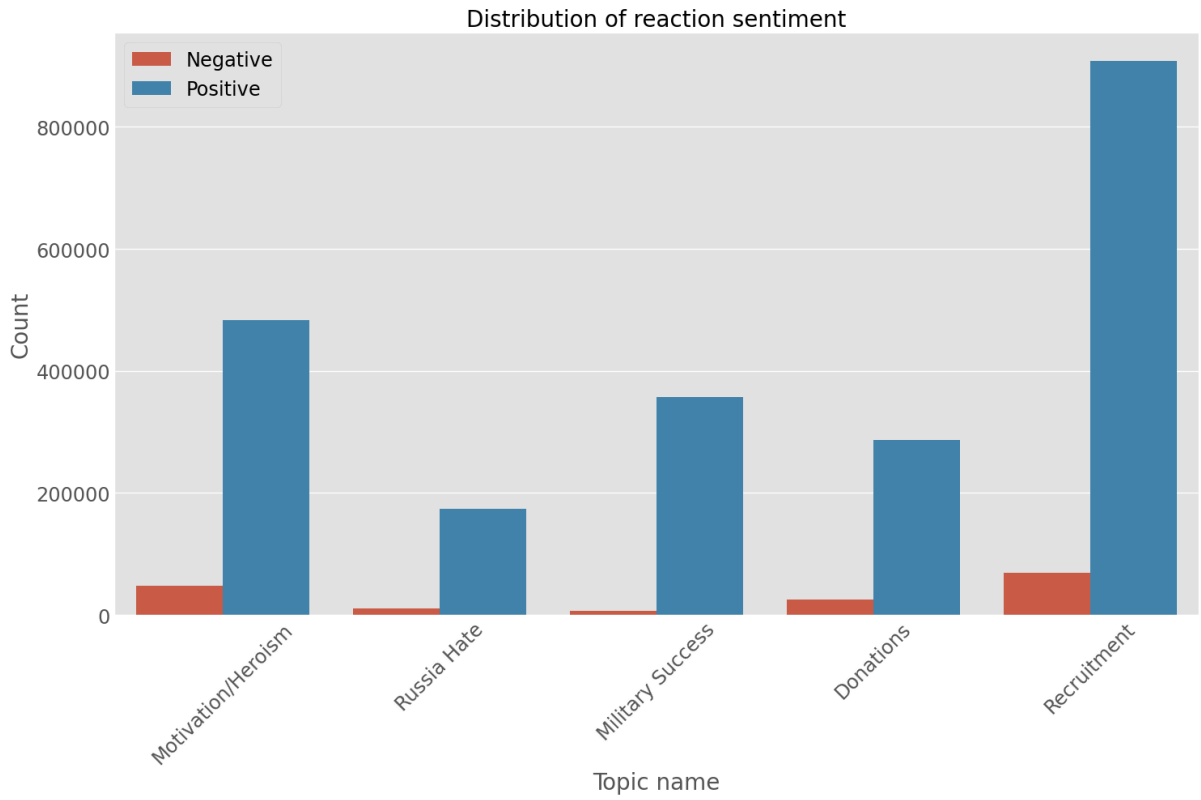
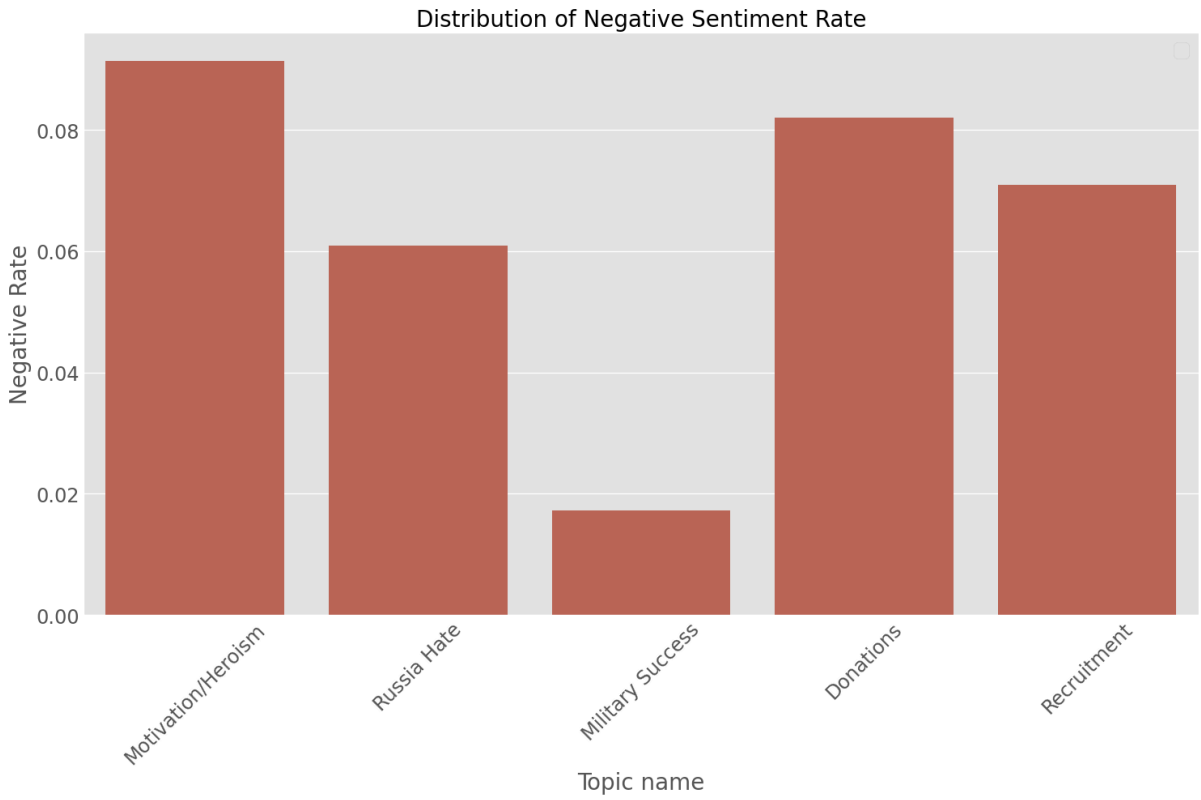


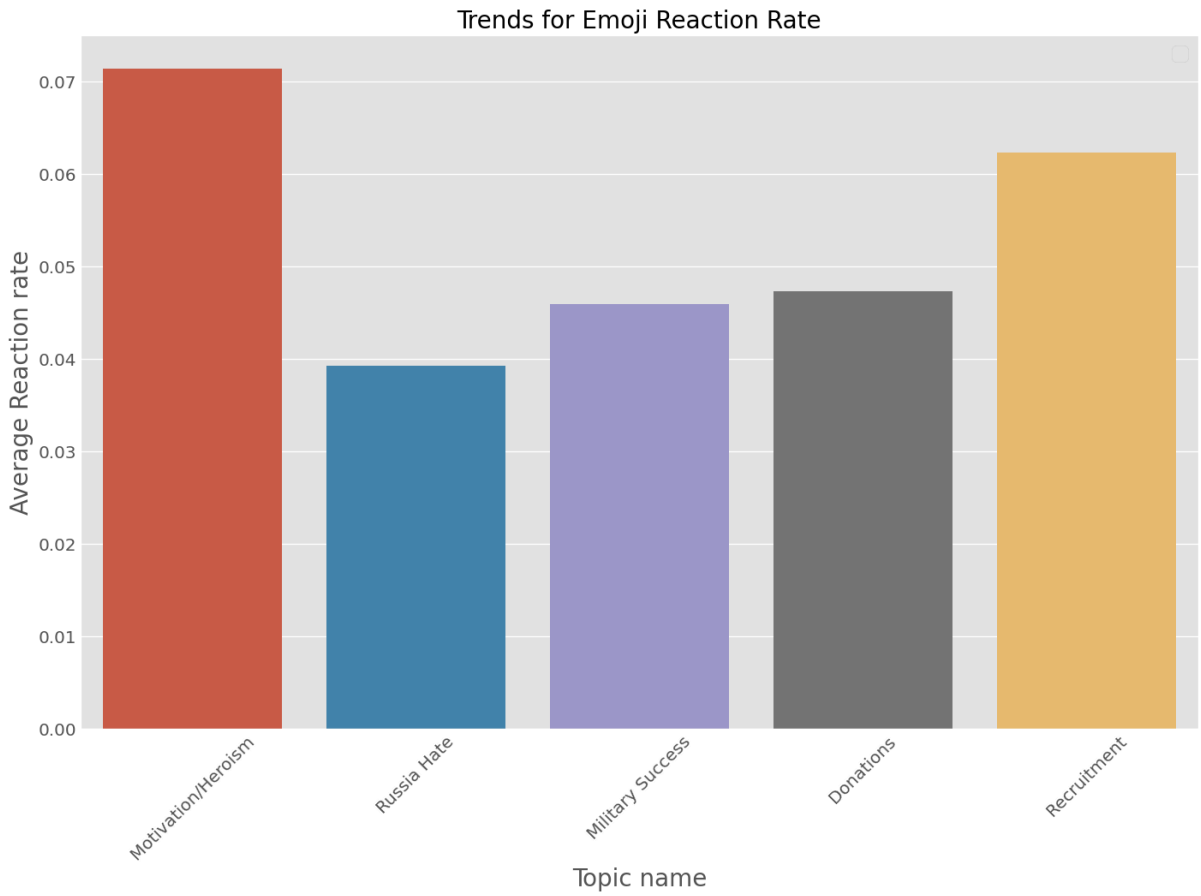
Figure 4: Negative sentiment rate per topic



Interaction rate (Nikolai)

As described in the methods section, emojis and views can also be used to estimate reaction rate. In Figure 5, we can see that documents in the "Motivation/Heroism" topic have the highest reaction rate, followed by the documents in the topics "Recruitment" and "Donations" which represent direct calls for support. The frames used in these three topics might therefore be more effective in mobilizing support, measured as the rate of user interaction. While all the topics have relatively high average reaction rates, directly asking for donations and calling for recruits likely converts into a higher number of tangible real world actions of support.

Figure 5: Average reaction rate for each topic

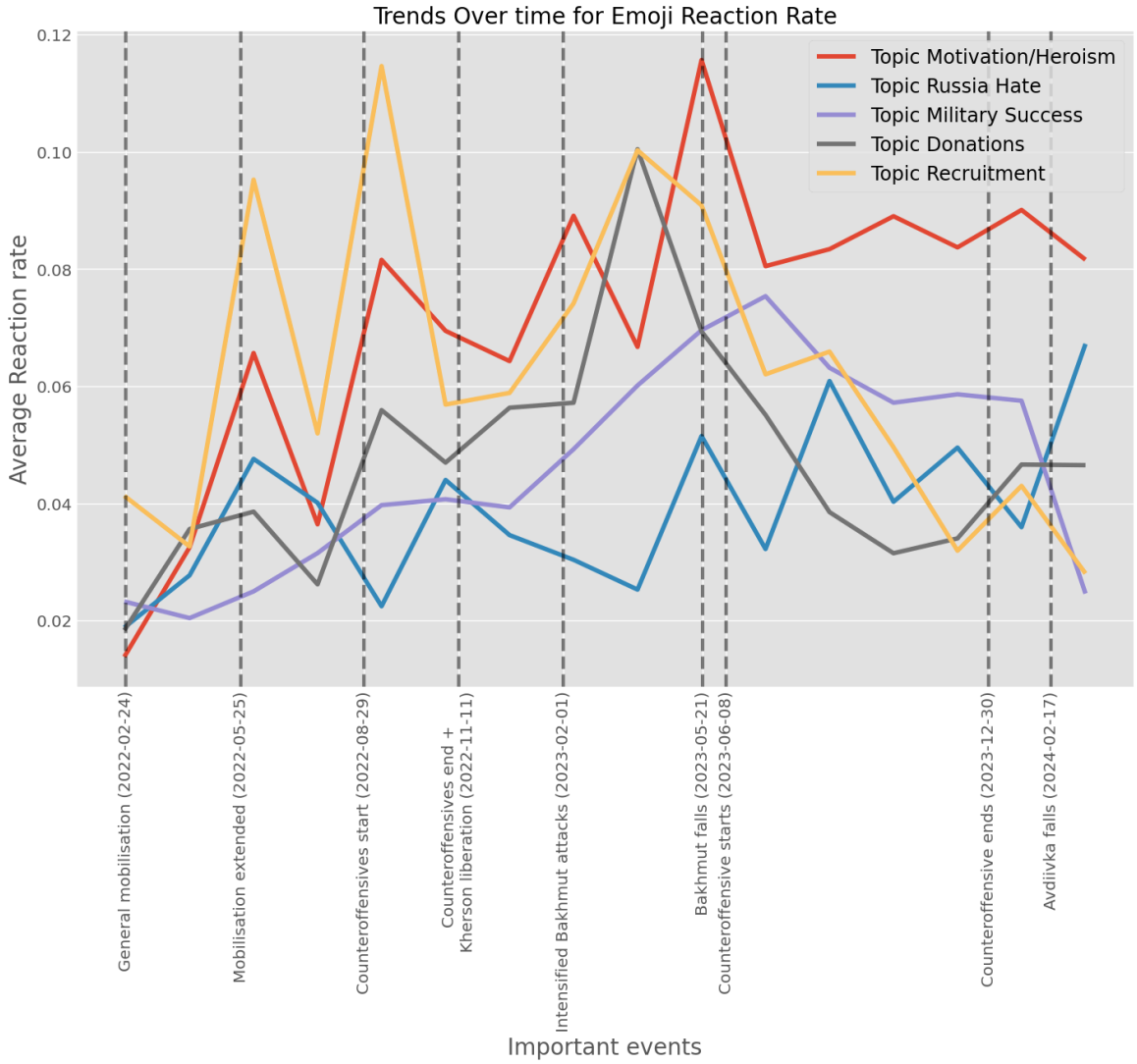


To analyze how the reaction rate developed over time, the period between 2022-02-24 and 2024-05-04 are binned into 16 date bins, and the average reaction rate is calculated for each topic in each time bin. Then, important events, and periods are plotted along with the data (see Figure 6).

The plot shows a spike in the average reaction rate for the topics “Recruitment”, “Russia Hate” and “Motivation/Heroism” in the months following the full scale invasion. For the “Recruitment” topic, this could likely be a reaction to the extension of the general mobilization which came into effect May 25 2022 (Interfax-Ukraine, 2022). This shows that Ukrainian Telegram users actively engage with especially military recruitment frames on Telegram during periods of general mobilization. Concurrent increases in the interaction rates with the other topics hint at general engagement with the war. A further increase in engagement is observable at the start of the Ukrainian counter offensives in the east and south, where Kharkiv and Kherson were liberated (Zafra et al., 2023). For “Recruitment” and “Motivation/Heroism” especially, this increase could be attributed to the successes in Kherson and Kharkiv.

In early 2023, during the most intense months of fighting in Bakhmut, the average reaction rate for messages in the topics “Motivation/Heroism”, “Recruitment” and “Donations” peaked (ISW, 2023). This indicates that military units framing of these topics were aligned with Telegram users, and thus able to increase reaction rates during a period of high casualty rates and increased munition expenditures. Interestingly, even with the fall of Bakhmut, “Military successes” continued to have an increasing reaction rate into the early stages of the 2023 counter offensive. Essentially this is an example of alignment in the framing of individual successful operations as military successes, independent of the overall state of the war. During the counteroffensive, reaction rates were lowered across topics, but especially for “Recruitment” and “Donations”. While the counteroffensive later has been described as slow and costly, the reaction rate for “Motivation/Heroism” was stable and at a high rate in this period. This indicates frame alignment between the military units and the Telegram users when met with high casualties. Subsequent to the fall of Avdiivka, engagement rates for several of the topics are trending downwards (Lukiv, 2024). This can be interpreted as a decrease in frame alignment across topic frames, or attributed to fatigue. Overall, mobilization of direct support as measured by reaction rate was high up until the fall of Bakhmut, after which this effort seems to have been less successful, barring “Motivation/Heroism.”.

Figure 6: Trends of average reaction rate for each topic. Important events are marked.



Discussion (Arvin)

In our qualitative analysis, we describe a mechanism where support is constructed in the topics of heroism, military success and Russia hate, and are consumed to align Ukrainian Telegram users with the need for donations and recruitment. Like in the qualitative analysis, the quantitative analysis also finds that the support "construction" topics both leverage positive and negative frames as measured by reaction sentiment. These topics might interact in the construction of a basis of support, meaning that the negative and anger mobilizing

frames are balanced by the positive frames of military success, in order to avoid generating a sense of despair that could undermine support. However, as measured by frequency, the topic "Motivation/Heroism" is the most frequent topic of the "construction" topics in most time periods. This disproportionality in the frequency of the "construction" topics may be problematic since the topic is made up largely of messages commemorating fallen soldiers, indicating high losses in the unit. While the promise of being commemorated as a hero may be a motivating factor when considering enlistment, the awareness of the extent of losses might also be a deterrent for recruitment. On the other hand, the documentation of casualties combined with the successful use of equipment could increase the motivation to donate and create alignment by communicating how the money can help prevent excessive losses.

In this paper, we have mostly looked at mobilization of support as a general phenomena covering all military units, rather than focusing on differences between the individual units. Looking at which units account for most messages within each topic, some units interestingly dominate "Donations" and "Recruitment". This could be attributed to the fact that some units such as Azov have more resources than others, possibly allowing them to employ social media teams for communication, which also would explain their state of the art recruitment websites. The higher engagement rates on recruitment posts from some of the military units can likely also be attributed to the fact that Ukrainians in the recruitment age bracket, have been known to preemptively sign up for popular units with established reputations. Given the better funding of these groups they are likely also able to better focus on recruitment rather than donations, or employ more effective strategies for donation campaigning.

Limitations

Translation (Nikolai)

Since our research is based on machine translated textual data, it is important to consider any possible downstream complications. The translated text is both used as a basis for our qualitative analysis but also as an input to BERTopic and the subsequent quantitative analysis. Given the corpus size, validating the quality of the full sample was not possible and we therefore only validated 15 random documents. Given our lack of proficiency in the Ukrainian language, we had to rely on using another translation, DeepL, to compare the translations. Therefore even though translations seemed coherent during this validation,

linguistic and semantic nuances could still have been lost or altered. On this issue, previous research found that Google Translate produced translations with high similarity to translations from a professional translator, while also giving similar topic outputs in bag-of-words topic models like LDA (De Vries et al., 2018). This is however not directly applicable to our research, since a bag-of-words approach was not utilized. Further, this article was written in 2018 and as such the results are from a previous version of Google Translate and therefore likely not reproducible. Despite these limitations, our translations as well as the topics produced appeared coherent based on our contextual knowledge on the war in Ukraine.

Data and sampling quality (Oliver)

While we expect that the vast majority of the people reacting and viewing the messages are Ukrainian, given the language of the channels, there could also be a number of international viewers. We expect it to mostly be a problem for messages associated with media content. In terms of the topic modeling approach, almost half of documents were classified as outliers due to topic reduction. Since the model is unsupervised we must assume that some documents were misclassified thus affecting the later analysis. Further, when sampling posts we disregarded the temporal aspect, and this approach emphasizes a global view of topics over the whole period in order to increase generalizability, but ignores the dynamic and temporal development of topics. Despite focusing on establishing generalizable topics and frames, our quantitative analysis is not generalizable given the non-causal approach. Instead we have discovered concepts which could serve as a basis for further research looking into the causal mechanics or further theoretical discovery of mobilization of support during wartime.

Alternative Approaches (Alex)

Although we have conducted the study according to our specific research design and utilizing the methods we find to be most appropriate, this is not necessarily the only way to go about examining our phenomenon of interest. In this section we want to take a look at other possible approaches to answer the question of how military units mobilize support through social media.

Firstly, the platform we chose to use as a datasite is not the only site where this phenomenon occurs and could perhaps be supplemented with other data sources. Most brigades have an online presence outside of Telegram, be it on Facebook, X, TikTok or their own website. The avenues for potential analyses based on text-as-data are many, but come with their own conditions for data collection. Access to data from Meta or X can be limited or behind a paywall, while the security measures behind military brigades' websites can differ, not to mention the ethical considerations coming into play. In any case, Telegram is not the only platform that generates data on the phenomenon and a bigger study could enrich the analysis with this data.

Visual communication has been left uncovered in our paper, but it plays a major role in the mobilization task of Ukrainian military units. A large part of the text messages we collected were supplemented with photo or video imagery, which could vary from showcases of soldiers' living conditions to more graphic footage from the battlefields. Furthermore, just visiting any brigade's website would make a lasting impression on anyone. During our immersion into the topic, we browsed through some of the websites and we were impressed by how visually engaging they were, to the point where we thought they resembled the webpages of modern shooter video games. This can be a potential object of future research efforts.

While we think the techniques we employed in our study are apt for our task, other techniques could possibly have yielded similar results while maintaining interpretability. For example, instead of running an unsupervised topic model, we could have used dictionary-based methods to outline relevant documents in our text corpus and substantiate the topics from our manual coding (Carlsen & Ralund, 2022).

Conclusion (All)

In this paper, we have attempted to answer the question of how Ukrainian military brigades mobilize support, namely through recruitment and donations. Through the digital traces of the military brigades' use of social media, we have been able to analyze this research question through a combination of qualitative and quantitative social data science methods. Through a digital ethnographic approach, we have been able to get a deeper understanding of both

where and how the military brigades interact on social media. By utilizing qualitative coding combined with unsupervised machine learning methods, we were able to extract five important topics together with their related documents from the whole corpus of Telegram messages from the military brigades. These topics were found to be *recruitment*, *donations*, *military success, Russian hate*, and *motivation/heroism*.

Through a qualitative framing analysis, based on theories from Erving Goffman, we were able to identify how the military brigades utilize different frames through these topics to mobilize support from the users of the Telegram, which were comparable to the concept of idiosyncrasy credit by Edwin Hollander (1958). We were able to expand on these insights through a quantitative analysis, where we investigated how the use of these topics have changed over time in the Telegram channels, which we could trace back to specific changes in the war with Russia. Furthermore, we analyzed the engagement these messages garnered from the brigades' user bases through the views they get, combined with the usage of emoji reactions, and how this relates to key events in the war. Lastly, our discussion of these findings has raised several points and alternative approaches that could be used for future research of the use of social media by military organizations.

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